

# WARAITIP LAOSANGPRATEEP

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Thailand / South Korea / Singapore / APAC · Available August 2026



Dual-degree engineer combining a Chemical Engineering foundation (Sirindhorn International Institute of Technology (SIIT), Thailand) with a Master's in Medical Engineering (University of Auckland, QS Top 100), underpinned by competitive international research experience across New Zealand and Taiwan. Brings a rare combination of rigorous process engineering fundamentals, hands-on precision materials characterisation, and applied data science, including machine learning model development and statistical validation, that translates directly into high-technology manufacturing, process engineering, and advanced materials roles across Singapore, Thailand, and the broader APAC region. Proven ability to design and execute systematic experimental programmes, derive actionable conclusions from complex datasets, and communicate technical findings with precision. Bilingual Thai-English with strong cross-cultural working experience.

## ■ EDUCATION

### Master of Medical Engineering [Referee](#)

Mar 2025 – Jun 2026

[University of Auckland](#) · [New Zealand](#) · GPA 7.00 / 9.00 (Graduated with Distinction)

- ▶ Thesis: Quantitative mechanical characterisation of neonatal lamb growth plate tissue via nanoindentation/microindentation, mapping stiffness and hardness across four structurally distinct zones (articular, hypertrophic, proliferative, resting)
- ▶ Applied statistical testing to isolate significant results from high-variance datasets, reconciling findings against published literature
- ▶ Managed the full experimental pipeline — cryogenic tissue prep, controlled conditions, data acquisition, and outlier analysis — to produce publication-quality figures

### Coursework Project — Machine Learning for Materials Characterisation (Python & Supervised Learning)

- ▶ Built and validated SVM classification models (Python, cross-validation) to classify meat types from spectral wavelength data
- ▶ Applied data preprocessing and model evaluation for materials and process data interpretation

### Coursework Project — Electric Vehicle Charging Infrastructure Business Case (Auckland Region)

- ▶ Developed a Single Stage Business Case (SSBC) for a regional EV charging programme, assessing strategic, financial, and operational feasibility
- ▶ Ran multi-criteria analysis against investment objectives: EV adoption, grid efficiency, transport integration, and equitable access
- ▶ Assessed delivery models, funding pathways, and risk for investment
- ▶ Coursework: Advanced Functional Materials · Battery & Superconductor · Semiconductor & Materials Science · Medical Device & Technology Development · Advanced Imaging Technologies (OCT, MRI) · Machine Learning for Biomedical Applications · Advanced Drug Disposition & Kinetics · Engineering Project Management · Medical Device Development & regulation · Project Business Case

### Bachelor of Engineering – Chemical Engineering

Aug 2020 – May 2024

[Thammasat University \(SIIT\)](#) · [Thailand](#) · [International Programme](#) · GPA 3.22 / 4.00

- ▶ Senior Project: Computational study of butylone- $\beta$ -cyclodextrin inclusion complex formation using molecular docking (AutoDock) and quantum chemical modelling (Gaussian/GaussView) (see PACCON 2024 poster below)
- ▶ Core engineering foundation: Chemical Reaction Engineering · Process Design · Plant Design · Thermodynamics · Heat & Mass Transfer · Fluid Mechanics · Environmental Engineering · Wastewater Treatment · Safety in Chemical Operations · Analytical and Instrumental (XRD, FTIR, UV-Vis, Raman, XRF, NMR, AAS, ICP-OES, SEM, TEM, TGA, BET, GC, GC-MS, HPLC, LC-MS)

## ■ RESEARCH & PROFESSIONAL EXPERIENCE

### Research Intern – Nanomaterial Characterisation (SWCNTs)

Jun 2023 – Jul 2023

Academia Sinica — Institute of Atomic and Molecular Sciences (IAMS-IIP) with Prof. Ching-Wei Lin · Taipei, Taiwan [Link](#)

- ▶ Selected for a competitive internship at a leading Asian materials science institute; characterised SWCNT defect populations via absorbance/fluorescence spectroscopy for cancer diagnostics
- ▶ Designed a DOE-style experimental matrix; processed multi-dimensional spectral data in Origin Lab, contributing findings to peer reports
- ▶ Achieved independent experimental operation within days of arrival, demonstrating rapid onboarding and cross-cultural adaptability

### Customer Service & Operations (concurrent with full-time postgraduate studies)

Aug 2024 – Jan 2026

Lily's Collection · Blue Elephant · Mi&Chi · Khao San Eatery · The Coffee Club · Auckland, NZ

- ▶ Balanced 5 concurrent part-time hospitality/retail roles while completing a full-time, research-intensive postgraduate degree, demonstrating reliability, time management, and disciplined prioritisation.
- ▶ Resolved day-to-day operational and customer issues independently across multiple fast-paced venues, requiring minimal supervision.

## ■ TECHNICAL SKILLS

**Process Engineering:** Chemical & process fundamentals; thermodynamics; heat & mass transfer; reaction engineering; fluid mechanics; process scale-up; P&ID awareness; Aspen Plus; HYSYS; HAZOP awareness

**Materials & Characterisation:** Microindentation; absorbance & fluorescence spectroscopy; nanomaterial characterisation; optical microscopy; mechanical property testing; thin-film material principles

**Semiconductor & Mfg:** Semiconductor manufacturing fundamentals; wafer fab process awareness; process control principles; yield improvement concepts; SPC; FMEA; CAPA; GMP; factory compliance

**Medical Device & Regulatory:** Medical device development lifecycle; design verification & validation; ISO 13485; IEC 60601-1; IEC 62366; ISO 14971; FMEA; regulatory documentation; technical report writing

**Machine Learning & Data:** Python (pandas, NumPy, matplotlib, seaborn, scikit-learn); SVM classification; cross-validation; supervised ML model development & validation; Origin Lab; statistical analysis

**Quality & Improvement:** DOE (Design of Experiments); SPC; Six Sigma (DMAIC); Lean/Kaizen; FMEA; Fault Tree Analysis; root cause analysis; ISO awareness

**Simulation & Computational:** Aspen Plus; Gaussian/GaussView; AutoDock; Discovery Studio Visualizer; molecular docking; computational chemistry; drug-material interaction modelling

**Biomedical Imaging:** OCT; MRI; Ultrasound — operating principles and data interpretation from dedicated postgraduate coursework

## ■ CERTIFICATIONS & ACHIEVEMENTS

- ▶ Microsoft Data Analysis with SQL, Excel & Power BI Specialisation (Coursera)
- ▶ KAIST Introduction to Semiconductor Process 1& 2 (Coursera)
- ▶ Google Data Analysis with Python Specialisation (Coursera)
- ▶ Poster Presentation – PACCON 2024 (Pure and Applied Chemistry International Conference)
- ▶ Academia Sinica IAMS-IIP Internship Certificate, Taiwan — international research programme

## ■ VOLUNTEER & LEADERSHIP

### Narrator & Lab Demonstrator — SIIT Open Days

Jul 2022 – May 2024

Thammasat University (SIIT) · Thailand

- ▶ Designed and facilitated live chemical engineering demonstrations for prospective students, building technical communication skills for stakeholder engagement

**Languages:** Thai (Native) · English (IELTS 6.5) · Korean (Elementary — Introductory, KEC NZ 2026; TOPIK I target 2027) · Japanese (Elementary)